FINRISK-AI

Unified AI/NLP Risk Engine & Downstream Portfolio Intelligence

The problem

Petabytes of unstructured text (news, filings, social) arrive every second.

Traditional risk relies on lagging, structured data and misses sentiment shifts.

Our solution

A real-time engine turns raw text into 3 machine-readable signals:

Sentiment Score · Event Class · Impact Severity

Downstream use

Module A: tactical index rebalancer (S&P 100 basket)

Module B: strategic stress test on a $1.25B bank portfolio

End-to-end system architecture

Three layers, one event bus

1 · Ingestion

News: GDELT, NewsAPI, Reuters/ Bloomberg headlines

Social: X cashtag feeds (AAPL, NVDA, TSLA)

→

2 · AI/NLP engine

Financial lexicon (Loughran-McDonald, FinBERT)

Negation & valence shifting

7-class event taxonomy, severity 1-10

NER & ticker mapping

→

3 · Execution

Low-latency event bus

Module A & Module B

REST API: /analyze, /rebalance, / stress-test

The Core Challenge: The AI/NLP Risk Engine

Every text becomes three structured signals

Sentiment S

Range -1.0 to +1.0, with confidence interval



Event class

- Geopolitical
- Macroeconomic
- Credit event
- M&A
- Product launch
- Regulatory action
- Earnings & guidance

Impact I

Scale 1.0 to 10.0 from event gravity, sentiment extremity and trigger words ("default", "blockade", "antitrust")



Impact ≥ 7.0 triggers automatic escalation to stress testing. Output is strict JSON for OMS/EMS.

Module A: tactical index rebalancer

10-stock mock S&P index reweighted by live sentiment

Allocation formula

W(raw) = W(base) + S × 0.45 × W(base)

Each weight stays between 2% and 25%; all weights sum to 100%.

Rebalance only when the change exceeds 0.5% (deadband).

AAPL	MSFT	NVDA	AMZN	GOOGL
META	TSLA	JPM	V	XOM

Sentiment in action

Bullish: NVDA AI datacenter launch → overweight

Bearish: TSLA delivery miss, AAPL antitrust remedies → underweight

+4.82%

cumulative alpha vs equal weight

1.84

Sharpe ratio

Module B: wholesale stress testing

Synthetic $1.25B portfolio, 10 tranches: loans, bonds, sovereign debt, derivatives, equity

Naval blockade

Equity -15%, credit spreads +180 bps, derivative MTM haircuts

Rate spike

Yields +225 bps, duration losses in fixed income

CRE default ($1.4B)

HY spreads +280 bps, loan defaults up to +6.5%

Portfolio value



CET1 capital ratio



Actions suggested: macro hedging swaptions and credit provisioning.

Engineering challenges & mitigations

Each risk has a built-in fix

Problem	Mitigation
Financial Polysemy & Semantic Inversion: general LLMs misinterpret financial slang (e.g., "crushed expectations" treated as negative).	FinBERT/Loughran-McDonald domain dictionary calibration with context windows.
Social Media Noise & Cashtag Spam: Twitter/X streams contain extreme noise, bot pump-and-dumps, and duplicate retweets.	Entity cross-verification, volume filtering, and exponential moving-average sentiment buffers.
Portfolio Concentration & Liquidity Constraints: unconstrained rebalancing creates high turnover and extreme concentration.	Bounded box constraints (2%-25% caps) and rebalancing threshold deadbands ($\Delta W > 0.5\%$).

Production Roadmap & Quantitative Results

What is built today and how it scales

Demonstrated deliverables

✓ Multi-source ingestion (news + X)

✓ NLP risk engine with 3 output metrics

✓ Module A rebalancer with live visualizer

✓ Module B stress tester with Basel III CET1

✓ REST API and risk_engine.py

Deployment path

1. Containerize as microservices (Docker/Kubernetes)
2. Stream backbone on Apache Kafka / Flink
3. Enterprise feeds: Bloomberg B-PIPE, FactSet